

$$\begin{array}{l} \nabla\times\\ \mathbf{E}=\frac{\partial\mathbf{B}}{\partial t},\nabla\times\\ \mathbf{H}=\frac{\partial\mathbf{D}}{\partial t}+\\ \mathbf{J},\mathbf{B}=\\ \mu\mathbf{H},\mathbf{D}=\\ \epsilon\mathbf{E}\\ \Delta t\leq\\ \frac{1}{c\sqrt{(1/\Delta x)^2+(1/\Delta y)^2+(1/\Delta z)^2}}\\ \Delta t\\ c\\ \Delta x\\ \Delta y\\ \Delta z\\ ?\\ ?\\ ?\\ ??\\ \hbar\\ 2.40\\ WS_2\\ \hbar\\ 4.0\\ 1/resolution\\ \hbar\\ 3.31\\ \hbar\\ t^{design}=\\ 0.257\,\mu m\\ 2\pi\\ \lambda=\\ 0.593\,\mu m\\ WS_2\\ S_x=\\ S_y=\\ W_{design}+\\ 2\,d_{pml}\\ W_{design}=\\ 4.5\,\mu m\\ d_{pml}=\\ 0.3\,\mu m\\ 50\,gridpoints/\mu m\\ N_x\times\\ N_y=\\ 226\times\\ 226\approx\\ 5.1\times\\ 10^4\\ \rho_j\in\\ [0,1]\\ W_{design}^2=\\ 20.25\,\mu m^2\\ ?\\ \lambda=\\ 0.593\,\mu m\\ E_g\approx\\ 2.26\,eV\\ w_0\approx\\ 800\,nm\\ \hat{r}_{peak}=\frac{1.35\lambda}{2}\\ w_0\sqrt{|\ell|/2}\\ 4.5\,\mu\\ \ell=\\ 05\\ 50\,pts/\mu\\ WS_2\\ WS_2\\ ,\\ {}_0\big[\hat{x}\,e^{-i\omega_0t}+i\,\hat{y}\,e^{-i\omega_0t}\big]\,f(t)\\ f(t)\\ \hat{x}\\ \hat{y}\\ 90^\circ\\ \sigma^+_\\ WS_2\\ ??\\ ?\\ \dot{\epsilon}_0\,\omega\,\sqrt{2\,Im(\mathbf{E}^*\cdot\mathbf{H})}\\ \epsilon_0=\\ \mu_0=\\ c=\\ 1\\ \omega=\\ 2\pi f\\ \pi f\,Im(\mathbf{E}^*\cdot\mathbf{H})\\ C>\\ 0\\ C<\\ 0\\ g\\ \hline C_{CPL,vac}\end{array}$$